MAT367
Intro to Differential Geometry

Class Lecture Notes

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

VI Differential Forms and Integration

12 Covectors and the 1-form

Lec 17 - Mar 12 (Week 9)

... are we reviewing?

to understand VF, need to understand them as:

- *coord-wise*: $X = \sum X_i \partial_i$, where X_i's are smooth fns (in any chart on M)
- *abstract*: a smooth section of the tangent bundle (smooth choice of tangent vec for each point)
- *functional* [or sheaf-theoretic?]: a derivation on $C^\infty(M)$
- *integral*: sth that generates a flow (a 1-param family of diffeos)

we can do sth similar for diff forms.

- *coord-wise*: k-form $\omega = \sum f_I \, \mathrm{d}\omega_{i_1} \wedge \cdots \wedge \mathrm{d}\omega_{i_k}$
- *abstract*: ω is a smooth choice of k-covec at each point (a section of some [the cotangent] vector bundle)
- *functional*: ω a kind of fn $(\mathfrak{X}(M))^k \longrightarrow C^\infty(M)$
- *integral*: a k-form is sth integrable over a k-mfld

Example 12.1

given $f \in C^\infty(M)$, defn its (exterior) differential $\mathrm{d}f : \mathfrak{X}(M) \longrightarrow C^\infty(M)$ by:

$$\mathrm{d}f(X) = X(f)$$

note, $\mathrm{d}f$ is $C^\infty(M)$-linear, i.e.:

$$\mathrm{d}f(gX + hY) = g\mathrm{d}f(X) + h\mathrm{d}f(Y)$$

that is an exercise.

Definition 12.2: Functional

a **1-form** on M is a $C^\infty(M)$-linear map $\mathfrak{X}(M) \longrightarrow C^\infty(M)$. the set ($C^\infty(M)$-alg) of 1-forms is denoted $\Omega^1(M)$.

hey, rmbr dual vspaces? given $\alpha \in V^*$, we can write $\alpha(v)$ as $\langle \alpha, v \rangle$. this emphasizes that if V finite-dim, $(V^*)^* = V$ canonically. that is, we can think of V as a hom $V^* \longrightarrow \mathbb{R}$. note $V^* \simeq V$, but not canonically unless we first fix an inner prod.

Definition 12.3

given basis $e_1, \ldots, e_n$ of V, the **dual basis** for V^* is $e^1, \ldots, e^n$, where $e^i(e_j) = \delta_{ij}$.

cotangent space is defnd precisely as u think. its elems are called **covectors**. in $T_p\mathbb{R}^n$, we have a distinguished basis $\partial_1, \ldots, \partial_n$. its dual basis is written $\mathrm{d}x_1, \ldots, \mathrm{d}x_n$, where:

$$\mathrm{d}x_i(a_1\partial_1 + \cdots + a_n\partial_n) = a_i$$

Definition 12.4

given a lin. map $F : V \longrightarrow W$, theres a **dual map** $F^* : W^* \longrightarrow V^*$ defn'd by $F^*\omega(v) = \omega(Fv)$.

given smooth map $f : M \longrightarrow N$, we have a linear map $T_pf : T_pM \longrightarrow T_{f(p)}N$ given by $v \mapsto f_*v = T_pf(v)$. therefore, $T_p^*f : T_p^*M \longrightarrow T_{f(p)}^*N$ is given as $v \mapsto f^*v = T_p^*f(v)$. the "base" map is as a pushforward, i.e. vecs are *covariant*. the dual is as a pullback, i.e. covecs are *contravariant*.

Example 12.5

given $f \in C^\infty(M)$, we can defn a covec $\mathrm{d}_pf \in T_p^*M$ by:

$$\mathrm{d}_pf(v) = v(f) = D_vf$$

in particular, in $\mathbb{R}^n$, we can write $\mathrm{d}_pf = \partial_{x_1}f\mathrm{d}x_1 + \cdots + \partial_{x_n}f\mathrm{d}x_n$. exercise... lol.

Proposition 12.6

given $f \in C^\infty(M, N), g \in C^\infty(N)$, we have $\mathrm{d}_p(f^*g) = f^*\mathrm{d}_p g$.

Proof.

for any $v \in T_pM$:

$$\begin{aligned}\langle \mathrm{d}_p(f^*g), v\rangle &= v(f^*g)\\ &= v(g \circ f)\\ &= (T_pf(v))(g)\\ &= \left\langle \mathrm{d}_{f(p)}g, T_pf(v)\right\rangle\\ &= \left\langle f^*\mathrm{d}_{f(p)}g, v\right\rangle\end{aligned}$$

note $\langle \cdot, \cdot \rangle$ is not an inner prod here. ■

Lec 18 - Mar 17 (Week 10)

last time, we talked abt covecs - this is enough for a "vibe" of a defn. what is a 1-form? it is a smooth choice of covec at each pt of a mfld.

Definition 12.7: Coordinate-wise

a map $p \mapsto \alpha_p$ sending pts to covecs is a **diff. 1-form** if for every chart (U, φ):

$$\alpha_{\varphi(p)} = \sum_{i=1}^{m} g_i(\varphi(p))\ \mathrm{d}x_i$$

where $g_i : \varphi(U) \longrightarrow \mathbb{R}$ is smooth.

for instance:

1) $f : M \longrightarrow \mathbb{R}$ as $\mathrm{d}f = \sum \partial_{x_i} f - \mathrm{d}x_i$ on a chart. as an exercise, show this is indep of choice of chart.

2) $y\mathrm{d}x$ on $\mathbb{R}^2$ is not $\mathrm{d}f$ for any f. to see this, note its partials don't commute [why?], or that $y\mathrm{d}x$ depends on y so f depends on y, so $\partial_y f \neq 0$. [what?]

Definition 12.8: Functional

a 1-form is a $C^\infty(M)$-linear $\mathfrak{X}(M) \longrightarrow C^\infty(M)$ given by applying the covec to pts.

Proposition 12.9

let α be a 1-form in the functional sense. then for any VF X, there exists a well-defn'd covec α_p w $p \in M$ s.t. $\alpha(X)(p) = \alpha_p(X_p)$.

Proof.

wts if $X_p = Y_p$, then $\alpha(X)(p) = \alpha(Y)(p)$. therefore, this gives a well-defn'd map $X_p \mapsto \alpha_p(X_p)$. that it's linear, ie a covec, is an exercise. by linearity, enough to show that if $X_p = 0$ then $\alpha(X)(p) = 0$.

indeed, if $X_p = 0$, write $X = \sum f_i Y_i$ where $f_i(p) = 0$ ($Y_i = \partial_i$ on a chart near p, then extend somehow). thus:

$$\alpha(X)(p) \;=\; \sum_i f_i(p)\alpha(Y_i)(p) \;=\; 0$$

■

Proposition 12.10

a) functional 1-forms can be restricted to a form on an open set $U \subseteq M$.

b) the two defns are equiv.

note that this is non-trivial since a VF on U is not always a restriction of a VF on M.

Exercise 12.11

state carefully what the above means, then prove it.

given $g : N \longrightarrow \mathbb{R}$, recall a pullback of g along $f : M \longrightarrow N$:

$$f^*g \;=\; g \circ f$$

for covecs, there is an induced map $T_pf : T_pM \longrightarrow T_{f(p)}N$ by $v \mapsto f_*v$, which in turn induces a map $T_p^*N \longrightarrow T_p^*M$ by $\nu \mapsto f^*\nu$, where ν is a covec.

now, given a 1-form $\omega \in \Omega^1(N)$, we can define its **pullback** along f by:

$$(f^*\omega)_p \;=\; f^*(\omega_{f(p)})$$

we claim this is indeed a 1-form, i.e. its a smooth choice. indeed, in coords, we have a smooth fn $f : U \longrightarrow V$ where $U, V \subseteq \mathbb{R}^m, \mathbb{R}^n$ resp. if $\omega \in \Omega^1(V)$ given by $\sum g_i \mathrm{d}x_i$, then T_pf has smooth coord fns $\partial_{x_j} f_i$. note:

$$f^*\big|_{f(p)} \;=\; (T_pf)^\intercal$$

so:

$$(f^*\omega)_p \;=\; f^*\Big|_{f(p)}(\omega_{f(p)}) \;=\; f^*\Big|_{f(p)} \sum g_i(f(p))\,\mathrm{d}x_i$$

these are all smooth in p, so we are done.

now, some properties of pullbacks:

- $(fg)^* \;=\; g^*f^*$
- if $g : M \longrightarrow \mathbb{R}$ smooth, $f^*(g\omega) \;=\; f^*g \cdot f^*\omega$
- $f^*(\mathrm{d}g) \;=\; \mathrm{d}(f^*g)$
- in coords:

$$\begin{aligned}(f^*\omega)_p \;=\; f^*\Big|_{f(p)} \sum_i g_i(f(p))\,\mathrm{d}x_i \;&=\; \sum_i g_i(f(p)) f^*\Big|_{f(p)} \mathrm{d}x_i \\ &=\; \sum_i \sum_j g_i(f(p)) \frac{\partial f_i}{\partial x_j}\,\mathrm{d}x_j\end{aligned}$$

Definition 12.12

the **cotangent bundle** of a mfld M is a mfld structure on:

$$T^*M \ := \bigsqcup_{p \in M} T_p^* M$$

with the canonical proj $\pi : T^*M \longrightarrow M$. given (U, φ), we get a chart on T^*M:

$$(\pi^{-1}(U), (\varphi^{-1})^* : \pi^{-1}U \longrightarrow \varphi(U) \times \mathbb{R}^m)$$

here, $\mathbb{R}^m$ is a copy of the cotangent space. note transition maps are smooth, as they are given by:

$$(\varphi \circ \psi^{-1})^* = (T_p(\varphi \circ \psi^{-1}))^\intercal \ : \ \varphi(U \cap V) \times \mathbb{R}^m \longrightarrow \psi(U \cap V) \times \mathbb{R}^m$$

this is just repackaging the coord-wise defn.

NOTE: with tangent bundles, $f : M \longrightarrow N$ defined $Tf : TM \longrightarrow TN$. here, we can only do sth similar when f is a diffeo, as pts go forward but vecs go back. OTOH, if $\omega \in \Omega^1(N)$ is defn'd as $s_\omega : N \longrightarrow T^*N$, then $f^*\omega$ is given by:

$$s_{f^*\omega}(p) \ = \ f^*\big|_{f(p)}(s_\omega \circ f(p))$$

Definition 12.13: Abstract

a 1-form is a section of the cotangent bundle

given $\gamma : I \longrightarrow M$, $\alpha \in \Omega^1(M)$, take $\gamma^*\alpha \in \Omega^1(I)$ as $\gamma^*(\alpha) = f(t)\ \mathrm{d}t$ for some $f(t)$.

with this, we can define the integral of a 1-form.

Definition 12.14

we define:

$$\int_\gamma \alpha \;=\; \int_a^b f(t)\ \mathrm{d}t$$

where $I = [a, b]$. we'll take smoothness on closed intervals in the extension way.

Proposition 12.15

if $\alpha = \mathrm{d}f$ for some $f : M \longrightarrow \mathbb{R}$, then:

$$\int_\gamma \alpha \;=\; f(\gamma(b)) - f(\gamma(a))$$

Proof.

we have that $\gamma^*\mathrm{d}f \;=\; \mathrm{d}(\gamma^* f) \;=\; \mathrm{d}(f \circ \gamma) \;=\; \partial_t(f \circ \gamma)\mathrm{d}t$. apply usual ftc. ■

for instance, $y\mathrm{d}x$ is not $\mathrm{d}f$, i.e. an *exact form*, since by integrating counterclockwise on $\partial[0, 1]^2$, we get:

$$\int_\gamma y\ \mathrm{d}x \;=\; 0 + 0 + \int_0^1 1(-\mathrm{d}t) + 0 \;=\; 1 \;\neq\; 0 \;=\; f(0, 0) - f(0, 0)$$

Proposition 12.16

integrating is invariant wrt reparametrization:

$$\int_\gamma \alpha \;=\; \pm \int_{\gamma\circ\kappa} \alpha$$

where $\kappa : [c, d] \longrightarrow [a, b]$ is a diffeo, and $\pm$ depends on orientation.

Proof.

we have:

$$\int_{\gamma\circ\kappa} \alpha \;=\; \int_c^d (\gamma\circ\kappa)^*\alpha \;=\; \int_c^d \kappa^*\gamma^*\alpha \;=\; \pm\int_{\kappa(c)}^{\kappa(d)} \gamma^*\alpha$$

after a substitution. ■

recall α is *exact* if its $\mathrm{d}f$ for some $f \in C^\infty$. in this case, $\int_\gamma \alpha$ depends only on endpts.

fact: this is an iff. if $\int_\gamma \alpha$ depends only on endpts, then defn $f(p) = \int_\gamma \alpha$ where γ is any curve p_0 to p (where we can choose p_0). to ensure f smooth and $\mathrm{d}f = \alpha$, do a coord calculation. ofc, this is iff $\int_\gamma \alpha = 0$ for γ a loop.

Example 12.17

let $M = S^1, \alpha = \mathrm{d}\theta$. then:

$$\int_{S^1} \mathrm{d}\theta \;=\; \int_\gamma \mathrm{d}\theta \;=\; 2\pi$$

where γ inj on (a,b) and $\gamma(a) = \gamma(b)$. thus, $\mathrm{d}\theta$ is not an exact form.

13 General k-forms

Lec 19 - Mar 19 (Week 10)

last time, we talked abt integrating 1-forms:

$$\int_\gamma \alpha \;=\; \int_a^b \gamma^*\alpha \;=\; \int_a^b \alpha_{\gamma(t)}(\gamma_*(\partial_t))\,\mathrm{d}t$$

this is indep of parametrization of γ, since $\alpha_{\gamma(t)}(rv) = r\alpha_{\gamma(t)}(v)$. integral remains the same by substitution. that $\alpha(X+Y) = \alpha(X) + \alpha(Y)$ doesnt matter for this computation.

we want k-forms to be sth we integrate over k-dim submflds. (in)formally, this is a map:

$$\omega : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \longrightarrow \mathbb{R}$$

then, for $F : U \subseteq \mathbb{R}^k \longrightarrow M$, we can define:

$$\int_{F(U)} \omega \;=\; \int_U \omega_{F(x)}(F_*\partial_1, \ldots, F_*\partial_k)\, \mathrm{d}x_1 \cdots \mathrm{d}x_k$$

we *want* invariance under reparametrization $G : U \longrightarrow V$ diffeo, that is:

$$= \int_{G(U)} \omega_{F(G^{-1}(y))} \left((F \circ G^{-1})_*\partial_1, \ldots, (F \circ G^{-1})_*\partial_k\right)\, \mathrm{d}y_1 \cdots \mathrm{d}y_k$$

which, by change of variables from 257, is given as:

$$= \pm \int_U \omega_{F(x)} \left((F \circ G^{-1})_*\partial_1, \ldots, (F \circ G^{-1})_*\partial_k\right) \left|\det TG^{-1}\right| \mathrm{d}x_1 \cdots \mathrm{d}x_k$$

so what do we acc need? recall the following fact: if V is k-dim'l, then given any map $\omega : V \times \cdots \times V \longrightarrow \mathbb{R}$ which is multilinear and skew-symmetric, then we have that $\omega(v_1, \ldots, v_k) = C \cdot \det[v_1 \;\cdots\; v_k]$. thus, if $\omega : V^k \longrightarrow \mathbb{R}$ for n-dim'l V is multilinear and alternating, its restriction to *any* k-dim subspace behaves like a det.

denoting these as alternating (or skew-sym) k-tensors, our vague defn is that a k-form $\omega \in \Omega^k(M)$ is a smooth choice of alt k-tensors $\omega_p : T_pM \times \cdots \times T_pM \longrightarrow \mathbb{R}$.

Definition 13.1: Functional

a **k-form** is an alternating multilinear map: $\omega : \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{k \text{ times}} \longrightarrow C^\infty(M)$.

for instance, $\omega \in \Omega^n(\mathbb{R}^n)$ always looks like:

$$\omega(X_1, \ldots, X_n)(p) \;=\; f(p) \det[(X_1)_p \;\cdots\; (X_n)_p]$$

for some $f \in C^\infty(\mathbb{R}^n)$. what do alt k-tensors look like in general?

Definition 13.2

let $I = \{i_1 < \cdots < i_k\} \subseteq \{1, \ldots, n\}$. define $\mathrm{d}x_I = \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}$ to be the alt k-tensor which satisfies:

$$\mathrm{d}x_I(\partial_{i_1}, \ldots, \partial_{i_k}) = 1 \qquad \mathrm{d}x_I(p_j, \text{ anything else }) = 0$$

where $j \notin I$. note this uniquely defns an alt k-tensor.

here, the $\wedge$ is simply a symbol (for now). intepretation: take $v_1, \ldots, v_k$ and compute $\det[\pi_I v_1 \ \cdots \ \pi_I v_k]$, where π_I projects to $\langle \partial_{i_1}, \ldots, \partial_{i_k} \rangle$.

Lemma 13.3

every alt k-tensor $\omega : (\mathbb{R}^n)^k \longrightarrow \mathbb{R}$ can be expressed uniquely as:

$$\omega = \sum_{\substack{I \subseteq \{1,\ldots,n\} \\ |I|=k}} \omega_I \, \mathrm{d}x_I \qquad \omega_I \in \mathbb{R}$$

Proof.

note $\omega_I = \omega(\partial_{i_1}, \ldots, \partial_{i_k})$, check that $\omega(v_1, \ldots, v_k)$ determined by these. (what?) ■

Proposition 13.4

any k-form on $U \subseteq \mathbb{R}^n$ is given by:

$$\omega = \sum_{\substack{I \subseteq \{1,\ldots,n\} \\ |I|=k}} f_I(p) \, \mathrm{d}x_I$$

for $f_I : U \longrightarrow \mathbb{R}$ smooth. any k-form on M can be written this way on charts. in particular, $\omega(X_1, \ldots, X_k)(p) = \omega_p((X_1)_p, \ldots, (X_k)_p)$ for some well-defined alt k-tensor ω_p.

pf is same as for VF and 1-forms.